$$C_{\mathrm{qd}}^{(1)}\mathrm{i1\_i2\_i3\_i4\_} \to$$

$$\frac{1}{16\pi^2}\ (-\frac{2}{243}\sum_p g_1^4\,\mathrm{LF}_{3,0}[m_{\tilde{d}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{5}{324}\sum_p g_1^4\,\mathrm{LF}_{4,-1}[m_{\tilde{d}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{8}{1215}\sum_p g_1^4$$

$$\mathrm{LF}_{5,-2}[m_{\tilde{d}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{2}{81}\sum_p g_1^4\,\mathrm{LF}_{3,0}[m_{\tilde{e}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{5}{108}\sum_p g_1^4\,\mathrm{LF}_{4,-1}[m_{\tilde{e}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{8}{405}\sum_p g_1^4\,\mathrm{LF}_{5,-2}[m_{\tilde{e}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}\sum_p g_1^4\,\mathrm{LF}_{3,0}[m_{\tilde{l}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{5}{216}\sum_p g_1^4\,\mathrm{LF}_{4,-1}[m_{\tilde{l}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{405}\sum_p g_1^4\,\mathrm{LF}_{5,-2}[m_{\tilde{l}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{243}\sum_p g_1^4\,\mathrm{LF}_{3,0}[m_{\tilde{q}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{5}{648}\sum_p g_1^4\,\mathrm{LF}_{4,-1}[m_{\tilde{q}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{4}{1215}\sum_p g_1^4\,\mathrm{LF}_{5,-2}[m_{\tilde{q}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{8}{243}\sum_p g_1^4\,\mathrm{LF}_{3,0}[m_{\tilde{u}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{5}{81}\sum_p g_1^4\,\mathrm{LF}_{4,-1}[m_{\tilde{u}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{32}{1215}\sum_p g_1^4\,\mathrm{LF}_{5,-2}[m_{\tilde{u}}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{81}\,g_1^4\,\mathrm{LF}_{3,0}[\tilde{\mu}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{54}\,g_1^4\,\mathrm{LF}_{4,-1}[\tilde{\mu}]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{8}{405}\,g_1^4\,\mathrm{LF}_{5,-2}[\tilde{\mu}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{972}\,g_1^4\,\mathrm{LF}_{2,1,0}[m_1,m_{\tilde{d}}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{972}\,g_1^4\,\mathrm{LF}_{2,2,-1}[m_1,m_{\tilde{d}}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{486}\,g_1^4\,\mathrm{LF}_{3,1,-1}[m_1,m_{\tilde{d}}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{972}\,g_1^4\,\mathrm{LF}_{4,1,-2}[m_1,m_{\tilde{d}}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{972}\,g_1^4\,\mathrm{LF}_{2,1,0}[m_1,m_{\tilde{d}}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{972}\,g_1^4\,\mathrm{LF}_{2,2,-1}[m_1,m_{\tilde{d}}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{486}\,g_1^4\,\mathrm{LF}_{3,1,-1}[m_1,m_{\tilde{d}}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{972}\,g_1^4\,\mathrm{LF}_{4,1,-2}[m_1,m_{\tilde{d}}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{3888}\,g_1^4\,\mathrm{LF}_{2,1,0}[m_1,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{3888}\,g_1^4\,\mathrm{LF}_{2,2,-1}[m_1,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{1944}\,g_1^4\,\mathrm{LF}_{3,1,-1}[m_1,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{3888}\,g_1^4\,\mathrm{LF}_{4,1,-2}[m_1,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{3888}\,g_1^4\,\mathrm{LF}_{2,1,0}[m_1,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{3888}\,g_1^4\,\mathrm{LF}_{2,2,-1}[m_1,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{1944}\,g_1^4\,\mathrm{LF}_{3,1,-1}[m_1,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{3888}\,g_1^4\,\mathrm{LF}_{4,1,-2}[m_1,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{144}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,1,0}[m_2,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{144}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,2,-1}[m_2,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{72}\,g_1^2\,g_2^2\,\mathrm{LF}_{3,1,-1}[m_2,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{144}\,g_1^2\,g_2^2\,\mathrm{LF}_{4,1,-2}[m_2,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{144}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,1,0}[m_2,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{144}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,2,-1}[m_2,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{72}\,g_1^2\,g_2^2\,\mathrm{LF}_{3,1,-1}[m_2,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{144}\,g_1^2\,g_2^2\,\mathrm{LF}_{4,1,-2}[m_2,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}[m_3,m_{\tilde{d}}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,2,-1}[m_3,m_{\tilde{d}}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}[m_3,m_{\tilde{d}}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{4,1,-2}[m_3,m_{\tilde{d}}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}[m_3,m_{\tilde{d}}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,2,-1}[m_3,m_{\tilde{d}}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}[m_3,m_{\tilde{d}}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{4,1,-2}[m_3,m_{\tilde{d}}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}[m_3,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,2,-1}[m_3,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}[m_3,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{4,1,-2}[m_3,m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}[m_3,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,2,-1}[m_3,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}[m_3,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{4,1,-2}[m_3,m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{27}\,g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,\mathrm{LF}_{2,1,0}[m_{\tilde{d}}^{p},\tilde{\mu}]\,\delta_{i3i4}+\frac{1}{54}\,g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,\mathrm{LF}_{2,2,-1}[m_{\tilde{d}}^{p},\tilde{\mu}]\,\delta_{i3i4}+$$

$$\frac{1}{54}\,g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,\mathrm{LF}_{3,1,-1}[m_{\tilde{d}}^{p},\tilde{\mu}]\,\delta_{i3i4}+\frac{1}{486}\,g_1^4\,\mathrm{LF}_{2,1,0}[m_{\tilde{d}}^{i3},m_1]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{972}\,g_1^4\,\mathrm{LF}_{3,1,-1}[m_{\tilde{d}}^{i3},m_1]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}[m_{\tilde{d}}^{i3},m_3]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}[m_{\tilde{d}}^{i3},m_3]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{486}\,g_1^4\,\mathrm{LF}_{2,1,0}[m_{\tilde{d}}^{i4},m_1]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{972}\,g_1^4\,\mathrm{LF}_{3,1,-1}[m_{\tilde{d}}^{i4},m_1]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}[m_{\tilde{d}}^{i4},m_3]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}[m_{\tilde{d}}^{i4},m_3]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{6}\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{a_e}^{pr}\,a_e^{pr}\,\mathrm{LF}_{3,1,0}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]+$$

$$\frac{1}{2}\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{a_e}^{pr}\,a_e^{pr}\,\mathrm{LF}_{4,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]-\frac{1}{3}\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{a_e}^{pr}\,a_e^{pr}\,\mathrm{LF}_{5,1,-2}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]-$$

$$\frac{1}{2}\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{a_d}^{pr}\,a_d^{pr}\,\mathrm{LF}_{3,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]+\frac{3}{2}\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{a_d}^{pr}\,a_d^{pr}\,\mathrm{LF}_{4,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]-$$

$$\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{a_d}^{pr}\,a_d^{pr}\,\mathrm{LF}_{5,1,-2}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]-\frac{1}{54}\,g_1^2\,\overline{y_d}^{pi3}\,y_d^{pi4}\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{p},\tilde{\mu}]\,\delta_{i1i2}+$$

$$\frac{1}{108}\,g_1^2\,\overline{y_d}^{pi3}\,y_d^{pi4}\,\mathrm{LF}_{2,2,-1}[m_{\tilde{q}}^{p},\tilde{\mu}]\,\delta_{i1i2}+\frac{1}{108}\,g_1^2\,\overline{y_d}^{pi3}\,y_d^{pi4}\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{p},\tilde{\mu}]\,\delta_{i1i2}+$$

$$\frac{1}{1944}\,g_1^4\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i1},m_1]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{3888}\,g_1^4\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i1},m_1]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{72}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i1},m_2]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{144}\,g_1^2\,g_2^2\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i1},m_2]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i1},m_3]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i1},m_3]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{1944}\,g_1^4\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i2},m_1]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{3888}\,g_1^4\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i2},m_1]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{72}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i2},m_2]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{144}\,g_1^2\,g_2^2\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i2},m_2]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{i2},m_3]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}\,g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{i2},m_3]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{27}\,g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,\mathrm{LF}_{2,1,0}[m_{\tilde{u}}^{p},\tilde{\mu}]\,\delta_{i3i4}-\frac{1}{27}\,g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,\mathrm{LF}_{2,2,-1}[m_{\tilde{u}}^{p},\tilde{\mu}]\,\delta_{i3i4}-$$

$$\frac{1}{27}\,g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,\mathrm{LF}_{3,1,-1}[m_{\tilde{u}}^{p},\tilde{\mu}]\,\delta_{i3i4}-\frac{1}{2}\,\tilde{\mu}^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{y_u}^{pr}\,y_u^{pr}\,\mathrm{LF}_{3,1,0}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]+$$

$$\frac{3}{2}\,\tilde{\mu}^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{y_u}^{pr}\,y_u^{pr}\,\mathrm{LF}_{4,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]-\tilde{\mu}^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\overline{y_u}^{pr}\,y_u^{pr}\,\mathrm{LF}_{5,1,-2}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]+$$

$$\frac{1}{3}\,g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\mathrm{LF}_{3,1,-1}[\tilde{\mu},m_1]-\frac{2}{3}\,g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\mathrm{LF}_{4,1,-2}[\tilde{\mu},m_1]+$$

$$\frac{1}{3}\,g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\mathrm{LF}_{5,1,-3}[\tilde{\mu},m_1]+g_2^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\mathrm{LF}_{3,1,-1}[\tilde{\mu},m_2]-$$

$$2\,g_2^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\mathrm{LF}_{4,1,-2}[\tilde{\mu},m_2]+g_2^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\mathrm{LF}_{5,1,-3}[\tilde{\mu},m_2]+$$

$$\frac{1}{54}\,g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,\mathrm{LF}_{2,1,0}[\tilde{\mu},m_{\tilde{d}}^{p}]\,\delta_{i3i4}+\frac{5}{108}\,g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,\mathrm{LF}_{3,1,-1}[\tilde{\mu},m_{\tilde{d}}^{p}]\,\delta_{i3i4}-$$

$$\frac{1}{108}\,g_1^2\,\overline{y_d}^{i2p}\,y_d^{i1p}\,\mathrm{LF}_{4,1,-2}[\tilde{\mu},m_{\tilde{d}}^{p}]\,\delta_{i3i4}+\frac{1}{108}\,g_1^2\,\overline{y_d}^{pi3}\,y_d^{pi4}\,\mathrm{LF}_{2,1,0}[\tilde{\mu},m_{\tilde{q}}^{p}]\,\delta_{i1i2}+$$

$$\frac{7}{108}\,g_1^2\,\overline{y_d}^{pi3}\,y_d^{pi4}\,\mathrm{LF}_{3,1,-1}[\tilde{\mu},m_{\tilde{q}}^{p}]\,\delta_{i1i2}-\frac{1}{54}\,g_1^2\,\overline{y_d}^{pi3}\,y_d^{pi4}\,\mathrm{LF}_{4,1,-2}[\tilde{\mu},m_{\tilde{q}}^{p}]\,\delta_{i1i2}-$$

$$\frac{1}{27}\,g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,\mathrm{LF}_{2,1,0}[\tilde{\mu},m_{\tilde{u}}^{p}]\,\delta_{i3i4}-\frac{1}{108}\,g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,\mathrm{LF}_{3,1,-1}[\tilde{\mu},m_{\tilde{u}}^{p}]\,\delta_{i3i4}-$$

$$\frac{1}{108}\,g_1^2\,\overline{y_u}^{i2p}\,y_u^{i1p}\,\mathrm{LF}_{4,1,-2}[\tilde{\mu},m_{\tilde{u}}^{p}]\,\delta_{i3i4}-\frac{1}{648}\,g_1^4\,\mathrm{LF}_{2,1,1,-1}[m_1,m_{\tilde{d}}^{i3},m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{324}\,g_1^4\,m_1^2\,\mathrm{LF}_{2,1,1,0}[m_1,m_{\tilde{d}}^{i3},m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{216}\,m_1\,g_1^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}$$

$$\mathrm{LF}_{2,2,1,-1}[m_1,m_{\tilde{d}}^{i3},m_{\tilde{q}}^{i2}]-\frac{1}{216}\,m_1\,g_1^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,\mathrm{LF}_{2,2,1,-1}[m_1,m_{\tilde{d}}^{i4},m_{\tilde{q}}^{i1}]-$$

$$\frac{1}{648}\,g_1^4\,\mathrm{LF}_{2,1,1,-1}[m_1,m_{\tilde{d}}^{i4},m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{324}\,g_1^4\,m_1^2\,\mathrm{LF}_{2,1,1,0}[m_1,m_{\tilde{d}}^{i4},m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{216}\,m_1\,g_1^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,\mathrm{LF}_{2,2,1,-1}[m_1,m_{\tilde{q}}^{i1},m_{\tilde{d}}^{i4}]-$$

$$\frac{1}{216}\,m_1\,g_1^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}\,\mathrm{LF}_{2,2,1,-1}[m_1,m_{\tilde{q}}^{i2},m_{\tilde{d}}^{i3}]-\frac{1}{36}\,g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\mathrm{LF}_{2,2,1,-2}[m_1,\tilde{\mu},m_{\tilde{d}}^{i3}]-$$

$$\frac{1}{36}\,g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\mathrm{LF}_{2,2,1,-2}[m_1,\tilde{\mu},m_{\tilde{d}}^{i4}]-\frac{1}{72}\,g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\mathrm{LF}_{2,2,1,-2}[m_1,\tilde{\mu},m_{\tilde{q}}^{i1}]-$$

$$\frac{1}{72}\,g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\mathrm{LF}_{2,2,1,-2}[m_1,\tilde{\mu},m_{\tilde{q}}^{i2}]-\frac{1}{8}\,g_2^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\mathrm{LF}_{2,2,1,-2}[m_2,\tilde{\mu},m_{\tilde{q}}^{i1}]-$$

$$\frac{1}{8}\,g_2^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\mathrm{LF}_{2,2,1,-2}[m_2,\tilde{\mu},m_{\tilde{q}}^{i2}]-\frac{1}{9}\,g_3^4\,\mathrm{LF}_{2,1,1,-1}[m_3,m_{\tilde{d}}^{i3},m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{2}{9}\,g_3^4\,m_3^2\,\mathrm{LF}_{2,1,1,0}[m_3,m_{\tilde{d}}^{i3},m_{\tilde{q}}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{9}\,m_3\,g_3^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}\,\mathrm{LF}_{2,2,1,-1}[m_3,m_{\tilde{d}}^{i3},m_{\tilde{q}}^{i2}]+$$

$$\frac{1}{9}\,m_3\,g_3^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,\mathrm{LF}_{2,2,1,-1}[m_3,m_{\tilde{d}}^{i4},m_{\tilde{q}}^{i1}]-\frac{1}{9}\,g_3^4\,\mathrm{LF}_{2,1,1,-1}[m_3,m_{\tilde{d}}^{i4},m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{2}{9}\,g_3^4\,m_3^2\,\mathrm{LF}_{2,1,1,0}[m_3,m_{\tilde{d}}^{i4},m_{\tilde{q}}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{9}\,m_3\,g_3^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,\mathrm{LF}_{2,2,1,-1}[m_3,m_{\tilde{q}}^{i1},m_{\tilde{d}}^{i4}]+$$

$$\frac{1}{9}\,m_3\,g_3^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}\,\mathrm{LF}_{2,2,1,-1}[m_3,m_{\tilde{q}}^{i2},m_{\tilde{d}}^{i3}]+\frac{1}{108}\,m_1\,g_1^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}$$

$$\mathrm{LF}_{2,1,1,0}[m_{\tilde{d}}^{i3},m_1,m_{\tilde{q}}^{i2}]-\frac{1}{108}\,m_1\,g_1^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{d}}^{i3},m_1,m_{\tilde{q}}^{i2}]-$$

$$\frac{2}{9}\,m_3\,g_3^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{d}}^{i3},m_3,m_{\tilde{q}}^{i2}]+\frac{2}{9}\,m_3\,g_3^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}$$

$$\mathrm{LF}_{3,1,1,-1}[m_{\tilde{d}}^{i3},m_3,m_{\tilde{q}}^{i2}]+\frac{1}{108}\,m_1\,g_1^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{d}}^{i4},m_1,m_{\tilde{q}}^{i1}]-$$

$$\frac{1}{108}\,m_1\,g_1^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{d}}^{i4},m_1,m_{\tilde{q}}^{i1}]-\frac{2}{9}\,m_3\,g_3^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}$$

$$\mathrm{LF}_{2,1,1,0}[m_{\tilde{d}}^{i4},m_3,m_{\tilde{q}}^{i1}]+\frac{2}{9}\,m_3\,g_3^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{d}}^{i4},m_3,m_{\tilde{q}}^{i1}]-$$

$$\frac{1}{12}\,\tilde{\mu}^2\,(\overline{y_d}^{pi3}\,y_d^{i1i4}\,\overline{y_u}^{i2r}\,y_u^{pr}+\overline{y_d}^{i2i3}\,y_d^{pi4}\,\overline{y_u}^{pr}\,y_u^{i1r})\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},\tilde{\mu}]+$$

$$\frac{1}{12}\,\tilde{\mu}^2\,(\overline{y_d}^{pi3}\,y_d^{i1i4}\,\overline{y_u}^{i2r}\,y_u^{pr}+\overline{y_d}^{i2i3}\,y_d^{pi4}\,\overline{y_u}^{pr}\,y_u^{i1r})\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},\tilde{\mu}]+$$

$$\frac{1}{24}\,\tilde{\mu}^2\,(\overline{y_d}^{pi3}\,y_d^{i1i4}\,\overline{y_u}^{i2r}\,y_u^{pr}+\overline{y_d}^{i2i3}\,y_d^{pi4}\,\overline{y_u}^{pr}\,y_u^{i1r})\,\mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{p},\tilde{\mu},m_{\tilde{u}}^{r}]+$$

$$\frac{1}{108}\,m_1\,g_1^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{i1},m_1,m_{\tilde{d}}^{i4}]-$$

$$\frac{1}{108}\,m_1\,g_1^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{i1},m_1,m_{\tilde{d}}^{i4}]-$$

$$\frac{2}{9}\,m_3\,g_3^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{i1},m_3,m_{\tilde{d}}^{i4}]+\frac{2}{9}\,m_3\,g_3^2\,\overline{y_d}^{i2i3}\,a_d^{i1i4}$$

$$\mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{i1},m_3,m_{\tilde{d}}^{i4}]+\frac{1}{108}\,m_1\,g_1^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{i2},m_1,m_{\tilde{d}}^{i3}]-$$

$$\frac{1}{108}\,m_1\,g_1^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{i2},m_1,m_{\tilde{d}}^{i3}]-\frac{2}{9}\,m_3\,g_3^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}$$

$$\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{i2},m_3,m_{\tilde{d}}^{i3}]+\frac{2}{9}\,m_3\,g_3^2\,y_d^{i1i4}\,\overline{a_d}^{i2i3}\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{i2},m_3,m_{\tilde{d}}^{i3}]-$$

$$\frac{1}{12}\,\tilde{\mu}^2\,(\overline{y_d}^{pi3}\,y_d^{i1i4}\,\overline{y_u}^{i2r}\,y_u^{pr}+\overline{y_d}^{i2i3}\,y_d^{pi4}\,\overline{y_u}^{pr}\,y_u^{i1r})\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},\tilde{\mu}]+$$

$$\frac{1}{12}\,\tilde{\mu}^2\,(\overline{y_d}^{pi3}\,y_d^{i1i4}\,\overline{y_u}^{i2r}\,y_u^{pr}+\overline{y_d}^{i2i3}\,y_d^{pi4}\,\overline{y_u}^{pr}\,y_u^{i1r})\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},\tilde{\mu}]+$$

$$\frac{1}{24}\,\tilde{\mu}^2\,(\overline{y_d}^{pi3}\,y_d^{i1i4}\,\overline{y_u}^{i2r}\,y_u^{pr}+\overline{y_d}^{i2i3}\,y_d^{pi4}\,\overline{y_u}^{pr}\,y_u^{i1r})\,\mathrm{LF}_{2,2,1,-1}[m_{\tilde{u}}^{r},\tilde{\mu},m_{\tilde{q}}^{p}]-$$

$$\frac{1}{4}\,\overline{y_d}^{ri3}\,\overline{y_d}^{i2p}\,y_d^{ri4}\,y_d^{i1p}\,\mathrm{LF}_{2,1,1,-1}[\tilde{\mu},m_{\tilde{d}}^{p},m_{\tilde{q}}^{r}]-$$

$$\frac{1}{2}\,\tilde{\mu}^2\,\overline{y_d}^{pi3}\,y_d^{pi4}\,\overline{y_u}^{i2r}\,y_u^{i1r}\,\mathrm{LF}_{2,1,1,0}[\tilde{\mu},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]+$$

$$\frac{1}{54}\,g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\mathrm{LF}_{1,1,1,1,-1}[m_1,m_{\tilde{d}}^{i3},m_{\tilde{d}}^{i4},\tilde{\mu}]+\frac{1}{108}\,g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}$$

$$\mathrm{LF}_{1,1,1,1,-1}[m_1,m_{\tilde{d}}^{i3},m_{\tilde{q}}^{i1},\tilde{\mu}]+\frac{1}{108}\,g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\mathrm{LF}_{1,1,1,1,-1}[m_1,m_{\tilde{d}}^{i4},m_{\tilde{q}}^{i2},\tilde{\mu}]+$$

$$\frac{1}{216}\,g_1^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\mathrm{LF}_{1,1,1,1,-1}[m_1,m_{\tilde{q}}^{i1},m_{\tilde{q}}^{i2},\tilde{\mu}]+\frac{1}{8}\,g_2^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}$$

$$\mathrm{LF}_{1,1,1,1,-1}[m_2,m_{\tilde{q}}^{i1},m_{\tilde{q}}^{i2},\tilde{\mu}]+\frac{2}{9}\,g_3^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\mathrm{LF}_{1,1,1,1,-1}[m_3,m_{\tilde{d}}^{i3},m_{\tilde{d}}^{i4},\tilde{\mu}]-$$

$$\frac{2}{9}\,g_3^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\mathrm{LF}_{1,1,1,1,-1}[m_3,m_{\tilde{d}}^{i3},m_{\tilde{q}}^{i1},\tilde{\mu}]-\frac{2}{9}\,g_3^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}$$

$$\mathrm{LF}_{1,1,1,1,-1}[m_3,m_{\tilde{d}}^{i4},m_{\tilde{q}}^{i2},\tilde{\mu}]+\frac{2}{9}\,g_3^2\,\overline{y_d}^{i2i3}\,y_d^{i1i4}\,\mathrm{LF}_{1,1,1,1,-1}[m_3,m_{\tilde{q}}^{i1},m_{\tilde{q}}^{i2},\tilde{\mu}])$$